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Masood3081

Enhance security hardening documentation for OpenWRT

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Security Hardening and Traffic Analysis

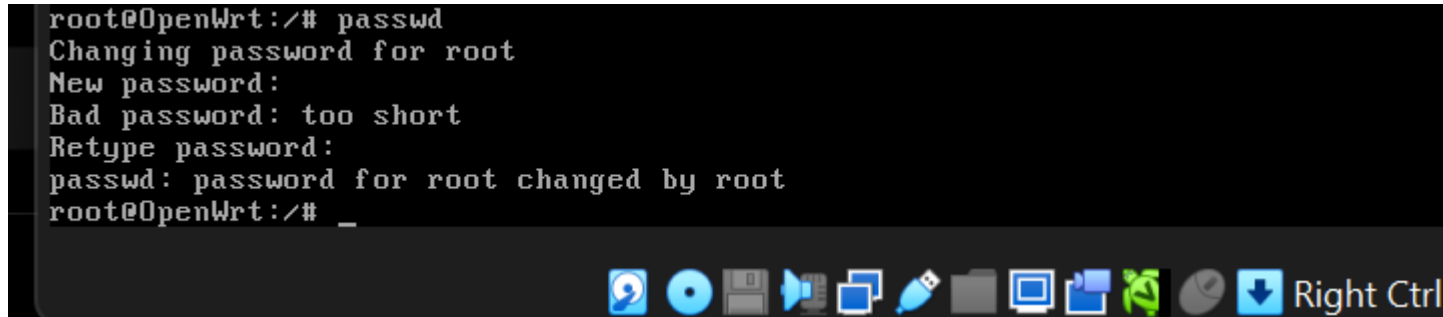
Harden the OpenWRT System

OpenWRT was hardened by changing the default root password, checking password storage, setting up SSH key-based login, and disabling an unnecessary service. These steps reduce the chance of unauthorised access and lower the attack surface of the router/firewall.

Change Default Root Password

The OpenWRT root password was changed using the `passwd` command.

```
root@OpenWrt:/# passwd
Changing password for root
New password:
Bad password: too short
Retype password:
passwd: password for root changed by root
root@OpenWrt:/# _
```

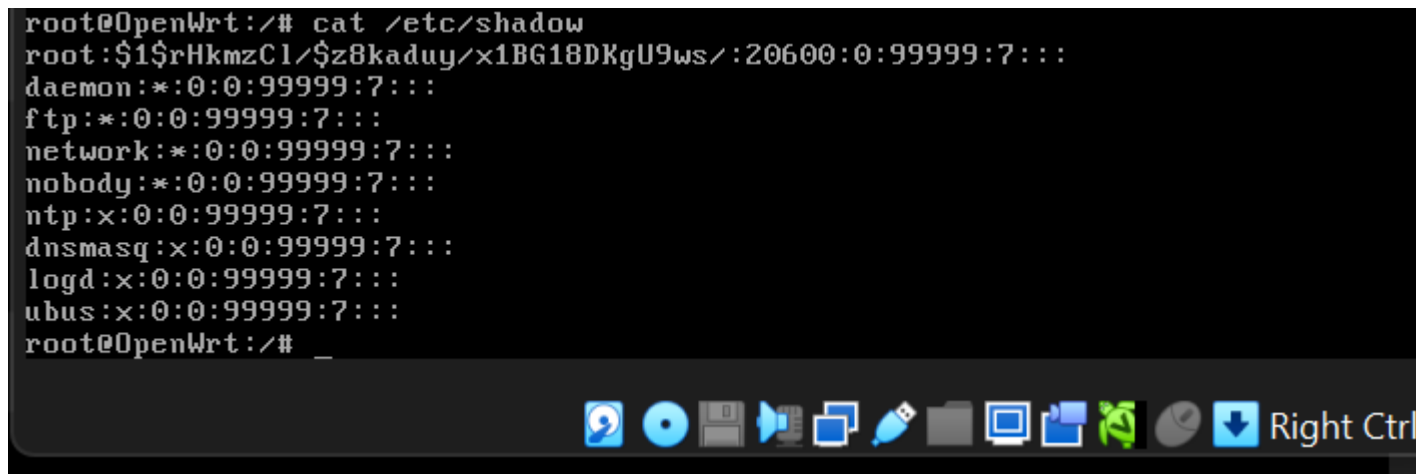
A terminal window with a dark background. The text shows a user at the root prompt running 'passwd' to change the root password. It prompts for a new password, rejects it as 'too short', asks to retype it, and then confirms the change. The prompt returns to root@OpenWrt:/#. At the bottom, there is a taskbar with various icons and the text 'Right Ctrl'.

Changing the default password is important because default or weak credentials can allow unauthorised users to access the router/firewall. A strong password reduces the risk of basic password guessing and unauthorised administrative login.

Examine Password Storage

The OpenWRT password storage file was checked using `/etc/shadow`.

```
root@OpenWrt:/# cat /etc/shadow
root:$1$rHkmzC1/$z8kaduy/x1BG18DKgU9ws/:20600:0:99999:7:::
daemon:*:0:0:99999:7:::
ftp:*:0:0:99999:7:::
network:*:0:0:99999:7:::
nobody:*:0:0:99999:7:::
ntp:x:0:0:99999:7:::
dnsmasq:x:0:0:99999:7:::
logd:x:0:0:99999:7:::
ubus:x:0:0:99999:7:::
root@OpenWrt:/# _
```

A terminal window with a dark background. The text shows a user at the root prompt running 'cat /etc/shadow' to view the password storage file. The output lists system users and their hashed passwords. The prompt returns to root@OpenWrt:/#. At the bottom, there is a taskbar with various icons and the text 'Right Ctrl'.

The password is stored as a hash rather than plaintext. This means the actual password is not directly visible in the file. Storing passwords as hashes reduces the impact if an attacker views the password file, because the attacker would still need to crack the hash to discover the original password.

Set Up SSH Key-Based Authentication

An SSH key pair was generated on the Windows host using the comment `keylab@user`.

```
The key fingerprint is:
SHA256:2uID/auDy6ZawpUT1H/im24g1D1e9Wif8atBmiFVeCc keylab@user
The key's randomart image is:
+--[ED25519 256]--+
|  ..          ..          |
|  .  .        o.E  .      |
|  .. o    ..+ o          |
|  .o. = o.o o          |
|  .+ .o S.....+        |
|  . ..o..=  . =o  .      |
|  ... .o+.+  o  .  .      |
|  o .o.o=.      ..        |
|  ...oo.==..  ..          |
+-----[SHA256]-----+
PS C:\Windows\system32> type $env:USERPROFILE\.ssh\id_ed25519_keylab.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJzFJ9iJY40p0MjbZOFhNoGublx5yzRGSxEioxOEcsbx keylab@user
PS C:\Windows\system32>
```

The public key was added to OpenWRT in the Dropbear `authorized_keys` file. SSH login was then tested successfully using the private key.

```
PS C:\Windows\system32> ssh -i $env:USERPROFILE\.ssh\id_ed25519_keylab root@192.168.56.2
```

BusyBox v1.35.0 (2023-01-03 00:24:21 UTC) built-in shell (ash)

```
- . . . W I R E L E S S F R E E D O M
```

OpenWrt 22.03.3, r20028-43d71ad93e

root@OpenWrt:~#

SSH key-based authentication is more secure than password-only authentication because access requires the private key. This reduces the risk of brute-force password attacks and improves administrative access security.

Disable Unnecessary Services

The available OpenWRT services were reviewed before disabling an unnecessary service.

```
root@OpenWrt:~# /etc/init.d/dropbear restart
root@OpenWrt:~# ls /etc/init.d
boot      firewall  odhcpd    system    urngd
cron      gpio_switch  rpcd      ucitrack
dnsmasq   led       sysctl    uhttpd
done      log       sysfixtime  umount
dropbear  network   sysntpd    urandom_seed
root@OpenWrt:~#
```



The `rpcd` service was disabled as hardening evidence. This service supports management-related remote procedure call functions. Disabling unnecessary management services reduces the attack surface.

```
root@OpenWrt:~# /etc/init.d/rpcd status
inactive
root@OpenWrt:~# /etc/init.d/rpcd disable
root@OpenWrt:~# /etc/init.d/rpcd stop
Command failed: Not found
root@OpenWrt:~# /etc/init.d/rpcd status
inactive
root@OpenWrt:~#
```



Core services such as `network`, `firewall`, `dropbear`, `uhttpd`, and `dnsmasq` were not disabled because they were needed for network connectivity, SSH access, firewall testing, and website hosting.

Traffic Analysis

Network traffic was captured using `tcpdump` on OpenWRT and analysed using Wireshark on the Windows host. Two captures were completed: one for HTTP traffic and one for SSH traffic.

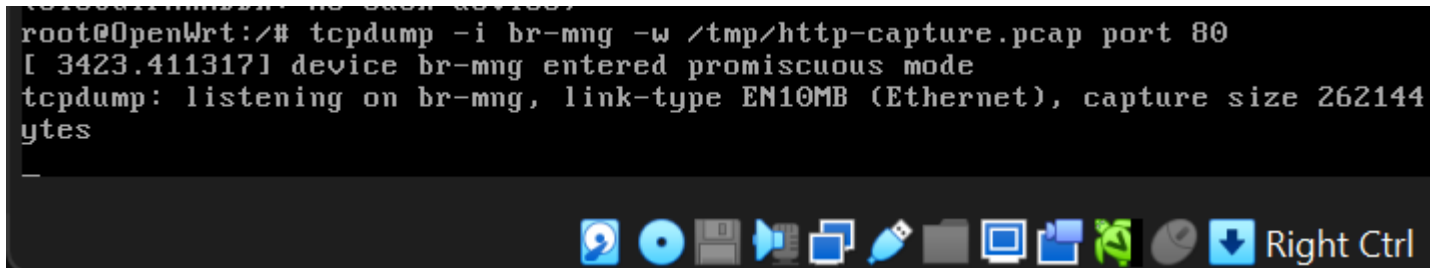
The packet capture files are included in the repository:

- [HTTP capture file](#)
- [SSH capture file](#)

HTTP Traffic Capture

HTTP traffic was captured while accessing the Sydney TaxCare Services test website from the Windows host browser.

```
root@OpenWrt:~# tcpdump -i br-mng -w /tmp/http-capture.pcap port 80
[ 3423.411317] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144
ytes
```

A screenshot of a terminal window with a black background and white text. The text shows the execution of the tcpdump command to capture traffic on port 80 from the br-mng interface. Below the terminal output, a portion of a Windows taskbar is visible, showing various icons like a globe, a CD, a folder, a USB drive, and a download arrow, along with the text 'Right Ctrl'.

The HTTP capture was opened in Wireshark and filtered using `http` .

The image shows a Wireshark packet capture window titled 'http-capture.pcap'. The packet list on the left shows a series of HTTP requests and 304 Not Modified responses. Packet 18 is selected, showing an HTTP GET request from 192.168.56.1 to 192.168.56.2. The packet details pane on the right shows the structure of the packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The Hypertext Transfer Protocol section shows the request line 'GET / HTTP/1.1' and various headers including Host, Connection, Cache-Control, Upgrade-Insecure-Requests, User-Agent, Accept, Accept-Encoding, Accept-Language, If-None-Match, and If-Modified-Since. The packet bytes pane on the right shows the raw data of the packet, with the request line 'GET / HTTP/1.1' highlighted in blue.

No.	Time	Source	Destination	Info	Dst port
9	0.000966	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
16	0.043861	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
18	0.569901	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
21	0.610826	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
23	0.736565	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
26	0.779030	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
28	0.859690	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
31	0.901079	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024

Frame 18: 592 bytes on wire (4736 bits), 592 bytes captured (4736 bits)

- Ethernet II, Src: 0a:00:27:00:00:0b (0a:00:27:00:00:0b), Dst: PCSSystemtec_4d:3c:5c
- Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.2
- Transmission Control Protocol, Src Port: 65024, Dst Port: 80, Seq: 539, Ack: 187, Len: 538
- Hypertext Transfer Protocol
 - GET / HTTP/1.1\r\n
 - Host: 192.168.56.2\r\n
 - Connection: keep-alive\r\n
 - Cache-Control: max-age=0\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n
 - If-None-Match: "62e-be0-6a173be7"\r\n
 - If-Modified-Since: Wed, 27 May 2026 18:45:59 GMT\r\n

Hypertext Transfer Protocol (http), 538 bytes

Packets: 32 · Displayed: 8 (25.0%)

Profile: Afiren

The HTTP request shows the Windows host `192.168.56.1` sending a `GET / HTTP/1.1` request to the OpenWRT web server at `192.168.56.2`.

http-capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Info	Dst port
9	0.000966	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
16	0.043861	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
18	0.569901	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
21	0.610826	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
23	0.736565	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
26	0.779030	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
28	0.859690	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
31	0.901079	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024

Frame 16: 189 bytes on wire (1512 bits), 189 bytes captured (1512 bits)

Ethernet II, Src: PCSSystemtec_4d:3c:5c (08:00:27:4d:3c:5c), Dst: 0a:00:27:00:00:0b (08:00:27:00:00:0b)

Internet Protocol Version 4, Src: 192.168.56.2, Dst: 192.168.56.1

Transmission Control Protocol, Src Port: 80, Dst Port: 65024, Seq: 52, Ack: 539, Len: 189

[2 Reassembled TCP Segments (186 bytes): #11(51), #16(135)]

Hypertext Transfer Protocol

HTTP/1.1 304 Not Modified\r\n

Connection: Keep-Alive\r\n

Keep-Alive: timeout=20\r\n

ETag: "62e-be0-6a173be7"\r\n

Last-Modified: Wed, 27 May 2026 18:45:59 GMT\r\n

Date: Wed, 27 May 2026 20:24:11 GMT\r\n

\r\n

[Request in frame: 9]

[Time since request: 0.042895000 seconds]

[Request URI: /]

[Full request URI: http://192.168.56.2/]

0000 0a 00 27 00 0b 08 00 27 4d 3c 5c 08 00 45 00 ...'M\...E...

0010 00 af b1 37 40 00 40 06 97 bd c0 a8 38 02 c0 a8 ...7@...8...

0020 38 01 00 50 fe 00 28 05 10 16 cb 4c 84 3e 50 18 8..P..(...L>P...

0030 0f a5 f1 f5 00 00 4b 65 65 70 2d 41 6c 69 76 65Ke ep-Alive

0040 3a 20 74 69 6d 65 6f 75 74 3d 32 30 0d 0a 45 54 : timeou t=20..ET

0050 61 67 3a 20 22 36 32 65 2d 62 65 30 2d 36 61 31 ag: "62e -be0-6a1

0060 37 33 62 65 37 22 0d 0a 4c 61 73 74 2d 4d 6f 64 73be7"... Last-Mod

0070 69 66 69 65 64 3a 20 57 65 64 2c 20 32 37 20 4d ified: W ed, 27 M

0080 61 79 20 32 30 32 36 20 31 38 3a 34 35 3a 35 39 ay 2026 18:45:59

0090 20 47 4d 54 0d 0a 44 61 74 65 3a 20 57 65 64 2c GMT..Da te: Wed,

00a0 20 32 37 20 4d 61 79 20 32 30 32 36 20 32 30 3a 27 May 2026 20:

00b0 32 34 3a 31 31 20 47 4d 54 0d 0a 0d 0a 24:11 GM T....

Frame (189 bytes) Reassembled TCP (186 bytes)

The HTTP response shows the OpenWRT web server replying to the Windows host. In this capture, the response includes HTTP status and header information.

http-capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Info	Dst port
9	0.000966	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
16	0.043861	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
18	0.569901	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
21	0.610826	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
23	0.736565	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
26	0.779030	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024
28	0.859690	192.168.56.1	192.168.56.2	GET / HTTP/1.1	80
31	0.901079	192.168.56.2	192.168.56.1	HTTP/1.1 304 Not Modified	65024

Frame 21: 189 bytes on wire (1512 bits), 189 bytes captured (1512 bits)

Ethernet II, Src: PCSSystemtec_4d:3c:5c (08:00:27:4d:3c:5c), Dst: 0a:00:27:00:00:0b (08:00:27:00:00:0b)

Internet Protocol Version 4, Src: 192.168.56.2, Dst: 192.168.56.1

Transmission Control Protocol, Src Port: 80, Dst Port: 65024, Seq: 238, Ack: 1077, Len: 189

[2 Reassembled TCP Segments (186 bytes): #19(51), #21(135)]

Hypertext Transfer Protocol

HTTP/1.1 304 Not Modified\r\n

Connection: Keep-Alive\r\n

Keep-Alive: timeout=20\r\n

ETag: "62e-be0-6a173be7"\r\n

Last-Modified: Wed, 27 May 2026 18:45:59 GMT\r\n

Date: Wed, 27 May 2026 20:24:12 GMT\r\n

\r\n

[Request in frame: 18]

[Time since request: 0.040925000 seconds]

[Request URI: /]

[Full request URI: http://192.168.56.2/]

The HTTP capture shows that HTTP traffic is readable in Wireshark. The request method, request URI, source IP address, destination IP address, response code, and headers are visible. This is a security risk because an attacker capturing HTTP traffic can learn information about the website request, server response, IP addresses, and other metadata. If full page content or sensitive information is transmitted over HTTP, it may also be visible in plaintext.

SSH Traffic Capture

SSH traffic was captured while performing an SSH session from the Windows host to OpenWRT.

```
root@OpenWrt:~# tcpdump -i br-mng -w /tmp/ssh-capture.pcap port 22
[ 4623.285148] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144
ytes
```

The SSH capture was opened in Wireshark and filtered using `ssh`.

ssh-capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ssh

No.	Time	Source	Destination	Info	Dst port
4	0.001287	192.168.56.1	192.168.56.2	Client: Protocol (SSH-2.0-OpenSSH_for_Windows_9.5)	
6	0.003007	192.168.56.2	192.168.56.1	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init	
7	0.003957	192.168.56.1	192.168.56.2	Client: Key Exchange Init	
9	0.048248	192.168.56.1	192.168.56.2	Client: Elliptic Curve Diffie-Hellman Key Exchange Init	
11	0.053334	192.168.56.2	192.168.56.1	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=1...	
12	0.056406	192.168.56.1	192.168.56.2	Client: New Keys	
14	0.099144	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=44)	
16	0.099998	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=44)	
17	0.100384	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=60)	
18	0.100939	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=52)	
19	0.101328	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=140)	
20	0.101845	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=100)	
21	0.104035	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=228)	
22	0.107492	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=36)	
23	0.110750	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=52)	
24	0.110881	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=44)	
25	0.111361	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=136)	
26	0.111590	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=36)	

Frame 4: 87 bytes on wire (696 bits), 87 bytes captured (696 bits)

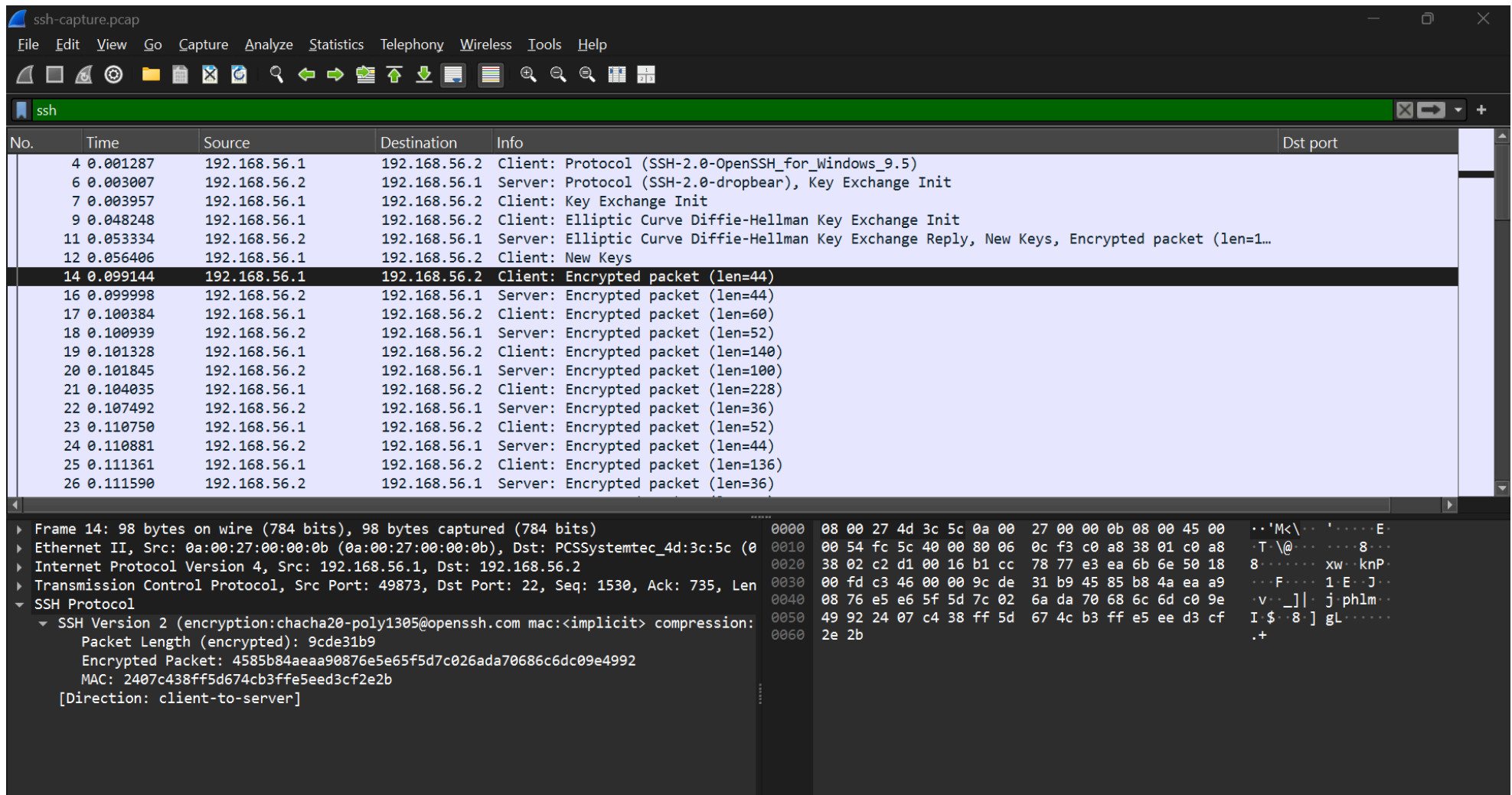
Ethernet II, Src: 0a:00:27:00:00:0b (0a:00:27:00:00:0b), Dst: PCSSystemtec_4d:3c:5c (08:00:27:00:4d:3c:5c)

Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.2

Transmission Control Protocol, Src Port: 49873, Dst Port: 22, Seq: 1, Ack: 1, Len: 33

SSH Protocol

The SSH packets show communication between the Windows host and OpenWRT. Unlike HTTP, the contents of the SSH session are not readable as plaintext.



The image shows a Wireshark capture of an SSH session. The top pane displays a list of packets with columns for No., Time, Source, Destination, Info, and Dst port. The bottom pane shows the details of the selected packet (Frame 14), including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and SSH Protocol. The SSH Protocol section indicates that the data is encrypted using chacha20-poly1305 and shows the packet length and MAC.

No.	Time	Source	Destination	Info	Dst port
4	0.001287	192.168.56.1	192.168.56.2	Client: Protocol (SSH-2.0-OpenSSH_for_Windows_9.5)	
6	0.003007	192.168.56.2	192.168.56.1	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init	
7	0.003957	192.168.56.1	192.168.56.2	Client: Key Exchange Init	
9	0.048248	192.168.56.1	192.168.56.2	Client: Elliptic Curve Diffie-Hellman Key Exchange Init	
11	0.053334	192.168.56.2	192.168.56.1	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=1...	
12	0.056406	192.168.56.1	192.168.56.2	Client: New Keys	
14	0.099144	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=44)	
16	0.099998	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=44)	
17	0.100384	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=60)	
18	0.100939	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=52)	
19	0.101328	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=140)	
20	0.101845	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=100)	
21	0.104035	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=228)	
22	0.107492	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=36)	
23	0.110750	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=52)	
24	0.110881	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=44)	
25	0.111361	192.168.56.1	192.168.56.2	Client: Encrypted packet (len=136)	
26	0.111590	192.168.56.2	192.168.56.1	Server: Encrypted packet (len=36)	

Frame 14: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)

- Ethernet II, Src: 0a:00:27:00:00:0b (0a:00:27:00:00:0b), Dst: PCSSystemtec_4d:3c:5c (08:00:27:4d:3c:5c)
- Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.2
- Transmission Control Protocol, Src Port: 49873, Dst Port: 22, Seq: 1530, Ack: 735, Len: 98
- SSH Protocol
 - SSH Version 2 (encryption:chacha20-poly1305@openssh.com mac:<implicit> compression:
 - Packet Length (encrypted): 9cde31b9
 - Encrypted Packet: 4585b84a90876e5e65f5d7c026ada70686c6dc09e4992
 - MAC: 2407c438ff5d674cb3ffe5eed3cf2e2b
 - [Direction: client-to-server]

The SSH capture shows encrypted SSH payload data. Commands entered during the SSH session are not visible in readable form. This demonstrates the benefit of encrypted protocols for protecting data in transit.

HTTP and SSH Comparison

HTTP traffic is unencrypted, so packet captures can show readable request and response information such as URLs, HTTP methods, headers, source IP addresses, and destination IP addresses. This can expose useful information to an attacker if the traffic is intercepted.

SSH traffic is encrypted, so Wireshark can show the source, destination, protocol, and packet flow, but not the readable session content. This protects administrative commands and session data from being viewed by someone capturing network traffic.

For this reason, encrypted protocols such as SSH are important for protecting sensitive data in transit, especially for administrative access to network devices such as routers and firewalls.